Practice Set 1 Paper 1 – Basic Sciences in Urology

Contents

1. Q1. Anatomy of the kidney; role of nephron-sparing surgery in the small renal mass (LAQ)
2. Q2. Anatomy of the prostate and its neurovascular bundles, with clinical details (LAQ)
3. Q3a. MR urography (SAQ)
4. Q3b. Urine cytology (SAQ)
5. Q3c. Varicocele: pathophysiology (SAQ)
6. Q3d. Anatomy of the renal collecting system and its relevance to PCNL (SAQ)
7. Q3e. Detrusor relaxants in OAB: compare and contrast (SAQ)
8. Q3f. Bilateral congenital VUR: pathophysiology and natural history (SAQ)

Practice paper compiled from the question bank.

Q1. Discuss anatomy of kidney. Describe role of nephron sparing nephrectomy in small renal mass.

Type: LAQ (25 marks) (also answered in 2017-Summer Paper 1, Q1; NSS also in 2022-Summer Paper 3, Q1 and 2025-Winter Paper 1, Q2)

"Nephron sparing nephrectomy" is read as nephron-sparing surgery (NSS): partial nephrectomy and ablation.

Surgical anatomy of the kidney

Position: T12–L3; right kidney 1–2 cm lower (liver). About 11 cm, 125–170 g. On the cone-shaped psoas, the upper pole is medial and posterior, and the kidney is angled 30–50° behind the coronal plane.

Relations

Aspect	Right	Left
Superior	Liver, adrenal, 12th rib	Spleen, stomach, adrenal, 11th–12th ribs
Medial/anterior	Duodenum (2nd part), hepatic flexure	Pancreatic tail with splenic vessels, splenic flexure, jejunum
Posterior (both)	Diaphragm, psoas, quadratus lumborum, subcostal nerve	Same

Coverings: true capsule, perirenal fat, **Gerota fascia**, pararenal fat. Gerota fascia is closed above and laterally but open below, so it contains tumour and collections, yet fluid tracks into the pelvis.

Internal structure: cortex, 8–18 pyramids; 0.4–1.2 million nephrons. Collecting system: see Q3d.

Vascular anatomy (the basis of partial nephrectomy)

- Hilum, anterior to posterior: vein, artery, pelvis, posterior segmental artery.
- Renal artery from the aorta at L1–L2; the right passes behind the IVC (inferior vena cava).
- **Five segmental end arteries**: posterior (first branch, behind the pelvis) and four from the anterior division (apical, upper, middle, lower). With no anastomoses, occlusion causes segmental infarction, but each segment can be clamped or resected separately.
- **Brödel line:** avascular plane between anterior (two-thirds) and posterior (one-third) territories, just behind the lateral border.
- **Accessory arteries** in 25–28%, end arteries too (see Practice Set 1 Paper 2, Q3c).
- **Veins anastomose freely**, so a segmental vein can be ligated safely. The long left renal vein receives adrenal, gonadal and lumbar veins.
- **Also know:** lymph to para-aortic (left) or interaortocaval and paracaval nodes (right).

Diagram to draw: transverse kidney section with segmental arteries and Brödel line.

Small renal mass (SRM)

- An enhancing renal mass **≤4 cm (cT1a)**, usually incidental.
- 20–30% of masses ≤4 cm and about 40% ≤2 cm are benign (oncocytoma, fat-poor angiomyolipoma); under 2% of the cancers metastasise.
- **Work-up:** multiphase CT (computed tomography) or MRI (magnetic resonance imaging), chest imaging, eGFR (estimated glomerular filtration rate), proteinuria. **RMB (renal mass biopsy)** if it may change management (non-diagnostic about 14%).
- **R.E.N.A.L. nephrometry** (Radius, Exophytic, Nearness to collecting system, Anterior/posterior, Location): 4–6 low, 7–9 moderate, 10–12 high complexity.

Why spare nephrons

- RN (radical nephrectomy) causes CKD (chronic kidney disease), linked to cardiovascular events and death. PN (partial nephrectomy) keeps about 90% of global function (RN 65–70%), with equal cancer control for cT1a.

Indications

Category	Situations
Imperative	Solitary kidney, bilateral tumours, RN would mean dialysis
Relative	Contralateral kidney threatened (stones, stenosis, diabetes); hereditary RCC (renal cell carcinoma)
Elective	cT1a with a normal contralateral kidney: **standard of care**

- **AUA (American Urological Association) 2021:** prioritise PN for cT1a. RN only if all three: high complexity; no CKD or proteinuria; normal contralateral kidney with eGFR likely >45 after RN.

Partial nephrectomy

1. Open, laparoscopic or robotic (equal oncology); early hilar control; intraoperative ultrasound.
2. Clamp the artery; ice slush if ischaemia ≥25–30 minutes is expected.
3. Excise with a thin rim; a negative margin matters, its width does not.
4. Close vessels and collecting system; renorrhaphy; drain.

- Function depends mainly on the **parenchyma preserved**; ischaemia is secondary (off-clamp gave no lasting gain in randomised trials).
- **Enucleation** along the pseudocapsule for familial, multifocal disease or severe CKD.
- **Trifecta:** negative margin, no major complication, warm ischaemia ≤25 minutes.

Other nephron-sparing options

- **Thermal ablation** (cryoablation, radiofrequency): alternative for cT1a **<3 cm**, percutaneous, biopsy first. Local recurrence is higher than PN but salvaged by repeat ablation.
- **Active surveillance:** <2 cm, elderly or comorbid; intervene if >3 cm or growth >5 mm a year (registry data: almost no metastases).

Complications of PN

- Urine leak (stent); **delayed haematuria from pseudoaneurysm or arteriovenous fistula** (angioembolisation); acute kidney injury; positive margin.
- **Hyperfiltration** if >50% of a solitary kidney is lost: proteinuria; ACE (angiotensin-converting enzyme) inhibitor.

Outcomes and evidence

- Elective PN: local recurrence 1–2%; imperative 3–5%.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** PN versus RN, ≤5 cm: better function, similar oncology; survival slightly favoured RN, but the trial is flawed.
- **Recent advances:** 3D (three-dimensional) models, segmental clamping.

Key points to remember:

- Five segmental end arteries; veins anastomose; Brödel line avascular.
- SRM ≤4 cm: 20–30% benign; PN is standard, ablation <3 cm, surveillance <2 cm.
- AUA: RN only if complex + no CKD + eGFR likely >45.
- Preserved parenchyma, not clamp time, governs function; EORTC 30904.

Q2. Anatomy of prostate and its neurovascular bundles with its clinical details.

Type: LAQ (25 marks) (nerve-sparing techniques also in 2023-Winter Paper 4, Q3a)

Introduction

- A fibromuscular, glandular organ around the proximal urethra, from bladder neck (**base**) to urogenital diaphragm (**apex**), in front of the rectum; about 20 g.
- Buds from the urogenital sinus at week 10, driven by androgen acting on the **mesenchymal** AR (androgen receptor).

Relations

- **Anterior:** retropubic space (of Retzius), puboprostatic ligaments, DVC (dorsal venous complex).
- **Posterior:** Denonvilliers fascia, then rectal ampulla, hence DRE (digital rectal examination) and TRUS (transrectal ultrasound).
- **Lateral:** levator ani with its fascia; NVB (neurovascular bundle) posterolaterally.
- **Base:** bladder neck; seminal vesicles behind. **Inside:** the urethra bends about 35° at the **verumontanum**, where the ejaculatory ducts open beside the utricle.

Zonal anatomy (McNeal 1981)

Zone	Glandular share	Position	Clinical
Peripheral (PZ)	About 70%	Posterior and lateral; longest ducts	**70% of cancers**; felt on DRE; biopsy target
Central (CZ)	About 25%	Cone around the ejaculatory ducts at the base	Few cancers; mimics HGPIN (high-grade prostatic intraepithelial neoplasia)
Transition (TZ)	About 5%	Two lobes beside the preprostatic sphincter; shortest ducts	**BPH (benign prostatic hyperplasia)**; anterior cancers (about 15%)
Periurethral glands	Small	Within the preprostatic sphincter	Middle-lobe BPH
Anterior fibromuscular stroma	No glands	Anterior	—

- **Capsule:** no true histological capsule, so cancer shows "extraprostatic extension". In BPH the compressed PZ forms the **surgical capsule**, the plane of enucleation. The capsule transmits growth pressure to the urethra, so TUIP (transurethral incision) relieves obstruction.

Fascial investments

Layer	Position	Surgical rule
Prostatic fascia	On the gland; carries DVC tributaries anteriorly	Stays on the prostate in nerve sparing
Levator fascia	Covers levator; fuses with prostatic fascia as the lateral pelvic fascia	Incised to reach the NVB
Denonvilliers fascia	Between prostate and rectum; thickest at base and vesicles	Excised completely for the posterior margin

Blood supply and lymphatics

- **Inferior vesical artery** ends as: **urethral group** (enters at the vesicoprostatic junction, 5 and 7 o'clock; supplies the adenoma) and **capsular group** (posterolateral, in the lateral pelvic fascia, marking the cavernous nerves).
- **Accessory pudendal arteries** run over the anterolateral prostate; dividing them reduces penile inflow.
- **Veins:** the deep dorsal vein passes under the pubis into a superficial branch and lateral (**Santorini**) plexuses, draining to the internal iliac vein. They anastomose freely, so tears bleed heavily.
- **Lymph:** obturator, internal and external iliac, presacral nodes.

Neurovascular bundle

- **Pelvic plexus:** parasympathetic S2–S4 plus sympathetic (T10–L2) via the hypogastric nerves; beside the rectum, centred on the seminal vesicle tip.
- **NVB of Walsh (Walsh and Donker, 1982):** capsular vessels plus **cavernous nerves**. The nerves are invisible; the vessels mark them.
- It runs posterolaterally **between prostatic and levator fascia**, 3–5 mm from the capsule, outside Denonvilliers fascia.
- At the apex it lies at **5 and 7 o'clock**, posterolateral to the sphincter. The nerves then swing to about **3 and 9 o'clock** at the mid-urethra and enter the corpora behind the dorsal penile artery. The apex is where they are most often injured.
- Distribution varies: posterolateral bundle, posterior "hammock", anterolateral fibres (basis of high anterior release).
- **Striated sphincter:** a vertical tube over the apex (Oelrich), pudendal nerve; passive continence.

Diagram to draw: axial prostate showing the three fasciae and NVBs at 5 and 7 o'clock.

Clinical details

- **BPH surgery:** TURP (transurethral resection of the prostate) stays proximal to the verumontanum (sphincter lies just distal); urethral-group vessels at 5 and 7 o'clock bleed.
- **Cancer:** PZ cancer exits posterolaterally along nerves; anterior TZ tumours are missed by DRE and biopsy, so MRI (magnetic resonance imaging) helps.
- **Radical prostatectomy (RP):**
  1. Incise endopelvic fascia; spare the pubourethral ligament.
  2. Ligate the DVC over the urethra before division.
  3. Preserve accessory pudendal arteries.
  4. Excise Denonvilliers fascia; the apex is the commonest positive-margin site.

Nerve-sparing plane	Course	Verdict
Interfascial	Between prostatic and levator fascia	**Standard plane**
Intrafascial	Under the prostatic fascia	Most nerves; more positive margins
Extrafascial	Outside levator fascia	Widest margin; ED (erectile dysfunction) likely

- **NVB injury:** incision, entrapment, **heat**, **traction**, so dissect athermally without traction. Excise an NVB for palpable extracapsular disease; margins are equal with or without nerve sparing.
- **Continence:** Retzius-sparing (Bocciardi 2010) and hood techniques preserve anterior support; continence returns earlier.

- **Also know:** rectal injury at the ampulla; PDE5 (phosphodiesterase type 5) inhibitor rehabilitation.

Key points to remember:

- PZ 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex most vulnerable.
- Interfascial plane is standard; avoid heat and traction.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.

Q3a. MR urography

Type: SAQ (10 marks)

Definition

- MRU (magnetic resonance urography): magnetic resonance imaging of kidneys, ureters and bladder giving anatomy and function **without ionising radiation**.

Techniques

	Static MRU	Excretory MRU
Sequence	Heavily T2-weighted; urine is bright	Gadolinium-enhanced T1
Contrast	None	GBCA (gadolinium-based contrast agent)
Depends on function	No	Yes; low-dose furosemide improves it
Best for	Dilated or obstructed system	Non-dilated system, excretion

- **Functional MRU (children):** hydration, bladder catheter, furosemide 1 mg/kg (maximum 20 mg), T2 images, then dynamic T1 after slow contrast injection. Patlak analysis gives differential function, GFR (glomerular filtration rate) and drainage.

Appearances

- Stones and calcification give **signal voids**; tumour, clot, gas and sloughed papillae are low-signal defects in bright urine.

Indications

- CT (computed tomography) urography contraindicated: contrast allergy, renal insufficiency, **pregnancy** (separates physiological from calculous dilatation; small stones missed).
- **Children:** complex anomalies, ectopic ureter in the incontinent girl, duplex systems, megaureter, crossing vessels in UPJ (ureteropelvic junction) obstruction, ectopic kidneys.
- Hydronephrosis: alternative to MAG3 (mercaptoacetyltriglycine) renography; equivalence not yet proven.
- Upper tract tumours and renal masses.

Advantages and limitations

Advantages	Limitations
No radiation; best anatomical detail	Cost, availability, long study (up to 1 hour)
Anatomy, function and drainage in one test	Sedation often needed under 6 years
As accurate as CTU (CT urography) for obstruction	Poor for stones; complex post-processing
Safe in iodinated contrast allergy	Pacemakers, metal implants, claustrophobia

Gadolinium safety

- **NSF (nephrogenic systemic fibrosis)** was the concern at GFR <30 mL/min/1.73 m². Group II agents (gadobenate, gadoteridol, gadoterate, gadobutrol) carry little or no risk even in end-stage disease.
- **Pregnancy:** gadolinium only if it clearly helps.
- **EAU (European Association of Urology):** CTU is preferred to MRU for upper tract urothelial cancer.

Diagram to draw: coronal static T2 MRU of a hydronephrotic kidney with a dark stone in the ureter.

Key points to remember:

- Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
- Stones are signal voids; CTU preferred for urothelial cancer.
- Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
- NSF risk minimal with group II agents.

Q3b. Urine cytology

Type: SAQ (10 marks)

Definition and principle

- Microscopy of shed urothelial cells (Papanicolaou, 1945).
- **Principle:** high-grade cells are poorly cohesive, shed readily and look atypical; low-grade cells look almost normal. So cytology detects **high-grade tumours and CIS (carcinoma in situ)** but misses low grade.

Specimens

- Voided urine (not the first morning sample, whose cells have degenerated), at least 30 mL; bladder washing; upper tract washings.

Performance

- Sensitivity 31–62%, specificity 94–100%; strongly grade-dependent.
- **False positives:** instrumentation, stones, infection, radiotherapy, intravesical BCG (bacille Calmette–Guérin), polyomavirus ("decoy" cells), ileal conduit.
- **False negatives:** low grade, low cellularity, degeneration.

The Paris System

- Published 2016; **2nd edition 2022**. Aim: detect **HGUC (high-grade urothelial carcinoma)**. Key criterion: **N:C (nuclear-to-cytoplasmic) ratio**, plus hyperchromasia, coarse chromatin and irregular nuclear membrane.

Category	Criteria
Non-diagnostic	Inadequate
Negative for HGUC	N:C <0.5, or atypia explained by a benign cause
AUC (atypical urothelial cells)	N:C ≥0.5 and one feature
Suspicious for HGUC	N:C ≥0.7, at least two features, few cells (<5–10)
HGUC	N:C ≥0.7, at least two features, many cells
LGUN (low-grade urothelial neoplasm)	Fibrovascular cores; rarely diagnosable
Other malignancy	Squamous, adenocarcinoma, secondary

- It reduces equivocal "atypia" reports; cancer risk rises with each category.

Clinical uses

- **Not** for screening or initial microhaematuria evaluation (AUA/SUFU [American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2020).
- Gross haematuria; NMIBC (non-muscle-invasive bladder cancer) surveillance (mandatory in high risk); suspected CIS; upper tract tumours.
- **Positive cytology, normal cystoscopy:** image upper tracts, selective washings ± ureteroscopy, random bladder and prostatic urethral biopsies.
- **Also know:** FISH (fluorescence in situ hybridisation; chromosomes 3, 7, 17, 9p21), NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), CxBladder: more sensitive, less specific; none replaces cystoscopy.

Diagram to draw: normal urothelial cell beside an HGUC cell with high N:C ratio.

Key points to remember:

- High specificity, low sensitivity; good for high grade and CIS.
- Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
- Positive cytology, normal cystoscopy: upper tracts, random biopsies.
- Not for screening or initial microhaematuria.

Q3c. Varicocele – Pathophysiology

Type: SAQ (10 marks)

Definition and epidemiology

- Abnormal dilatation and tortuosity of the **pampiniform plexus** of the internal spermatic veins.
- About 15% of men, 35% with primary and 75–81% with secondary infertility. Appears after age 10, peaks at Tanner 3. About 90% left-sided.
- **Venous anatomy:** anterior (internal spermatic), middle (deferential) and posterior (cremasteric) groups; the last two give return after repair and cause recurrence.

Why it develops

1. **Left-sided anatomy:** the left vein enters the left renal vein at a right angle, 8–10 cm higher than the right enters the IVC (inferior vena cava). The taller column overloads its valves. The right vein enters obliquely, which acts as a valve.
2. **Valve incompetence** or absence; continuous reflux predicts progression.
3. **Nutcracker:** left renal vein compressed between aorta and SMA (superior mesenteric artery).
4. **Collateral pathways**; **genetic** (4–8 times commoner in first-degree relatives); tall, thin habitus.

- **Secondary:** a **right-sided or non-decompressing** varicocele suggests a retroperitoneal or renal mass, so image the abdomen.

How it damages the testis

Mechanism	Explanation
Heat (primary)	Lost countercurrent heat exchange; the testis normally runs 2–4 °C below core
Oxidative stress	Raised reactive oxygen species → sperm DNA fragmentation
Hypoperfusion and stasis	Hypoxia, toxin accumulation
Apoptosis	Germ cell loss
Leydig dysfunction	Lower testosterone than controls

- The injury is **progressive and duration-dependent**, and may affect **both** testes.

Consequences

- **Testicular hypotrophy** (asymmetry above about 20% guides adolescent repair); poor semen; hypogonadism; ache on standing (2–11%).
- Repair improves semen, DNA fragmentation, testosterone and pregnancy (**Madgar RCT [randomised controlled trial], 1995:** 44% versus 10% at 1 year).

Grading

Grade	Finding
Subclinical	Doppler only (not treated)
I	Palpable with Valsalva only
II	Palpable, not visible
III	Visible

Diagram to draw: left and right testicular veins draining into renal vein and IVC.

Key points to remember:

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage.
- Right or non-decompressing varicocele: image for a renal mass.
- Injury is progressive; subclinical varicocele is not treated.

Q3d. Anatomy of renal collecting system and its relevance to PCNL

Type: SAQ (10 marks) (also answered in 2022-Summer Paper 1, Q1 and 2017-Winter Paper 1, Q1)

Anatomy of the collecting system

- **Minor calyx:** a cup around a papilla, with side projections (**fornices**); it drains by an infundibulum to a major calyx and pelvis. **Compound calyces** cluster at the poles.
- 7–9 calyces (upper 3, interpolar 3–4, lower 2–3); hilar ones in **anterior and posterior rows**.
- **Pelvis:** 3–10 mL, intrarenal or extrarenal; a **duplex pelvis** does not link upper and lower calyces.
- The kidney is rotated 30–50° posteriorly, so **posterior calyces point towards the prone surgeon**.

Sampaio classification

Type	Frequency	Midzone drainage
A-I	45%	Via a polar group
A-II	17.2%	**Crossed calyces** to both groups
B-I	21.4%	Independent major calyx
B-II	16.4%	Minor calyces straight into the pelvis

- **Perpendicular minor calyx (11.4%):** hidden on pyelography.

Vascular relations

- Interlobar arteries run in the **columns of Bertin**, hugging the infundibula; the posterior segmental artery passes behind the pelvis.
- **Brödel line:** avascular plane behind the lateral border, along the posterior calyces.

Puncture site (Sampaio)	Vascular injury
Upper-pole infundibulum	67% (arterial 26%)
Mid-pole infundibulum	Arterial 23%
Lower-pole infundibulum	Infundibular artery in 38%
Calyceal fornix	Venous <8%, **no arterial injury**

Relevance to PCNL (percutaneous nephrolithotomy)

- **Target:** a posterior calyx with a short, wide infundibulum, entered **through the papilla and fornix** along its axis, never through the infundibulum or pelvis.
- **Identify posterior calyces:** they shorten when the C-arm rotates towards the surgeon; air fills them when prone.
- **Upper pole:** best reach to pelvis, UPJ (ureteropelvic junction) and staghorns; chest risk above the 12th rib, highest above the 11th.
- **Lower pole:** chest-safe; retrorenal colon risk (left); poor UPJ reach.
- **Also know:** traps: perpendicular or crossed calyces, duplex pelvis.

Diagram to draw: transverse kidney section with a needle through a posterior fornix.

Key points to remember:

- Sampaio A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
- Fornix puncture: no arterial injury; upper-pole infundibulum 67%.
- Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.

Q3e. Compare and contrast various detrusor relaxants in management of OAB

Type: SAQ (10 marks) (mirabegron also answered in 2024-Winter Paper 3, Q3d)

Introduction and mechanisms

- In OAB (overactive bladder), drugs follow behavioural therapy. Two classes: **antimuscarinics** and **β3-adrenoceptor agonists**.
- **Antimuscarinics:** acetylcholine acts on **M3** receptors (raising intracellular calcium) to contract the detrusor; the commoner M2 enhances it. Blockade reduces involuntary contractions and afferent signalling.
- **β3 agonists:** β3 is the commonest detrusor adrenoceptor. Activation raises **cAMP (cyclic adenosine monophosphate)** and relaxes the detrusor during storage, sparing the voiding contraction.

Agents

Drug	Distinguishing feature	Usual dose
Oxybutynin	Non-selective; direct relaxant; metabolite causes dry mouth	5 mg two to three times daily; extended-release (ER), patch
Tolterodine	Bladder over salivary selective	ER 4 mg daily
Fesoterodine	Prodrug of tolterodine's active metabolite	4–8 mg daily
Solifenacin	10-fold M3 > M2; long half-life	5–10 mg daily
Darifenacin	About 30-fold M3-selective	7.5–15 mg daily
Trospium	Quaternary; does not cross the blood–brain barrier	20 mg twice daily
β3: mirabegron	Inhibits CYP2D6 (a liver enzyme): metoprolol interaction	25–50 mg daily
β3: vibegron	No CYP2D6 interaction	75 mg daily

Comparison

Feature	Antimuscarinics	β3 agonists
Efficacy	Similar	Similar (TAURUS: mirabegron ≈ tolterodine)
Dry mouth, constipation	Common; main reason for stopping	Near placebo
Cognition	Central M1 blockade; **dementia risk 10–65% higher with use >3 months**	No anticholinergic burden
Retention	Up to 3% in men	Low
Contraindications	Urinary or gastric retention, uncontrolled narrow-angle glaucoma	Uncontrolled hypertension (≥180/110 mmHg)

- **Choice:** in the elderly, prefer a β3 agonist or ER trospium, darifenacin or fesoterodine; ER forms cause less dry mouth.

Combination

- Solifenacin plus mirabegron beats each alone (**SYNERGY**); in non-responders it equals solifenacin 10 mg with less dry mouth (**BESIDE**). In men, add to an α-blocker; watch PVR (post-void residual).

- **Also know:** intravesical oxybutynin; imipramine; refractory OAB: onabotulinumtoxinA 100 U, neuromodulation.

Diagram to draw: detrusor cell: acetylcholine → M3 → contraction; noradrenaline → β3 → cAMP → relaxation.

> **Key points to remember:**
> - M3 mediates contraction; β3 raises cAMP to relax.
> - Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
> - Solifenacin + mirabegron (SYNERGY) works.

Q3f. B/l congenital VUR - Pathophysiology and natural history

Type: SAQ (10 marks) (VUR management also answered in 2015-Summer Paper 3, Q2 and 2020-Summer Paper 1, Q2)

"B/l congenital VUR" is read as bilateral primary (congenital) VUR (vesicoureteral reflux), often detected antenatally.

Pathophysiology

- **Embryology:** a ureteric bud arising too early over-rotates, so the orifice lies **high and lateral** with a short tunnel. Its abnormal contact with the metanephros causes **renal dysplasia**, so high-grade VUR and dysplasia coexist.
- **Failed UVJ (ureterovesical junction) flap valve:** filling normally compresses the intramural ureter against detrusor backing. Tunnel length to diameter is **5:1** normally versus 1.4:1 in refluxing units (Paquin). High infant male voiding pressure and BBD (bladder and bowel dysfunction) add a secondary component.

Renal damage	Reflux-associated dysplasia	Acquired scar
Cause	Abnormal bud	Infected urine plus VUR
Typical patient	Boy, grade IV–V, antenatal hydronephrosis	Girl, recurrent febrile UTI (urinary tract infection), BBD
Preventable	No	Yes

- **Intrarenal reflux** through compound polar papillae needs only 2 mmHg in neonates, so scars are polar. **Sterile reflux rarely scars** unless pressure is high.
- **Also know:** massive bilateral VUR gives the megacystis–megaureter association.

International grading: I ureter only; II up to calyces, undilated; III mild–moderate dilatation; IV moderate dilatation and tortuosity; V gross dilatation, papillary impressions lost.

Natural history

- Resolution comes from **elongation of the intravesical ureter** with growth. Grade II resolves in 63–85%, grade III in about 50%; **grade IV–V and bilateral grade III rarely** (International Reflux Study).
- **Bilaterality and older age delay resolution:** persistence is highest in bilateral grade IV, and in bilateral grade III at 6–10 years. Infants with grade IV–V may still resolve.
- **Consequences of bilateral damage:** adult hypertension in 34–38% of scarred children; CKD (chronic kidney disease), with reflux nephropathy a leading paediatric transplant diagnosis; poor growth; pre-eclampsia (worse with bilateral scars). Correction cannot reverse dysplasia.

Diagram to draw: long oblique tunnel versus short lateral one.

> **Key points to remember:**
> - One embryological fault: lateral orifice, short tunnel, dysplasia.
> - Dysplasia (boys, congenital) versus scar (girls, infection).
> - Grade IV–V and bilateral grade III rarely resolve.
> - Scarred children: 34–38% adult hypertension.

Rapid Revision Sheet – Practice Set 1 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss anatomy of kidney. Describe role of nephron sparing nephrectomy in small renal mass. (LAQ)

- Five segmental end arteries; veins anastomose; Brödel line avascular.
- SRM ≤4 cm: 20–30% benign; PN is standard, ablation <3 cm, surveillance <2 cm.
- AUA: RN only if complex + no CKD + eGFR likely >45.
- Preserved parenchyma, not clamp time, governs function; EORTC 30904.

Q2. Anatomy of prostate and its neurovascular bundles with its clinical details. (LAQ)

- PZ 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex most vulnerable.
- Interfascial plane is standard; avoid heat and traction.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.

Q3a. MR urography (SAQ)

- Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
- Stones are signal voids; CTU preferred for urothelial cancer.
- Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
- NSF risk minimal with group II agents.

Q3b. Urine cytology (SAQ)

- High specificity, low sensitivity; good for high grade and CIS.
- Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
- Positive cytology, normal cystoscopy: upper tracts, random biopsies.
- Not for screening or initial microhaematuria.

Q3c. Varicocele – Pathophysiology (SAQ)

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage.
- Right or non-decompressing varicocele: image for a renal mass.
- Injury is progressive; subclinical varicocele is not treated.

Q3d. Anatomy of renal collecting system and its relevance to PCNL (SAQ)

- Sampaio A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
- Fornix puncture: no arterial injury; upper-pole infundibulum 67%.
- Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.

Q3e. Compare and contrast various detrusor relaxants in management of OAB (SAQ)

- M3 mediates contraction; β3 raises cAMP to relax.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Solifenacin + mirabegron (SYNERGY) works.

Q3f. B/l congenital VUR - Pathophysiology and natural history (SAQ)

- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Dysplasia (boys, congenital) versus scar (girls, infection).
- Grade IV–V and bilateral grade III rarely resolve.
- Scarred children: 34–38% adult hypertension.